



KFSZ

Fully Stabilized Zirconia

KEY PROPERTIES

- High temperature resistance
- Oxygen ion conductivity
- Superior corrosion resistance
- High thermal insulation capacity
- Good chemical stability

APPLICATIONS

- Active membranes
- Oxygen sensors (partial pressure measurement)
- Electrochemical devices

MAIN PROPERTIES

Property	Value	Unit
Density	5.7	g/cm ³
Flexural strength	180	MPa
Hardness	14	GPa
Fracture Toughness	4.0	MPa·√m
Young Modulus	260	GPa
Thermal Conductivity	3	W/m·K
CTE (20–1000°C)	10.5	x10 ⁻⁶ K ⁻¹

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)		2000 °C
Recomended temp		<1800 °C
Continuous temp		1700 °C

ELECTRICAL BEHAVIOR

Ionic conductor	
Conductivity type	Oxygen ion conduction

MATERIAL INFORMATION

Material type	Fully stabilized zirconia
Structure	Ionic conductive ceramic
Key feature	Oxygen ion transport
Machinability	Medium
Color	White

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

Nanoker Research S.L.

Polígono de Olloniego, Parcela 21A, Nave 4-8, 33660, Oviedo (Asturias), Spain

(+34) 985 20 76 13 | info@nanoker.com